

Anthony Hughes

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I am a PhD student working on privacy and security in language models, and prior to my PhD I spent ten years in industry as a software engineer. My PhD focuses on developing algorithms for protecting private information. From this I learnt about pre-, mid- and post-training methods as well as circuit-based interpretability techniques to understand how models memorise and surface sensitive information.

I am now directing all my efforts to AI safety. As a SPAR fellow, I built a benchmark and methods for detecting whether models have been backdoored through poisoned data. This now includes scenarios where pretraining data is poisoned to backdoor AI oversight models, work which is being funded by a BlueDot grant. Additionally, my independent work examines vulnerabilities in safety-critical components such as classifiers and probes.

To me, what is both interesting and important is attacking a defence. I am interested in work that requires a deep understanding of the adversary and the defender. My aim is to be part of building robust monitoring and a better science of model behaviour. A technical safety research fellowship is the natural next step toward doing this full time.

Education

- **PhD Computer Science**
 - University of Sheffield, 2024-*Current (estimated completion Jan 2027)*
 - Additional Modules: Signal and Speech Processing, Reinforcement Learning, Time-series Analysis
- **MSc Computational Linguistics (Distinction)**
 - University of Wolverhampton, 2021-2023
 - Modules: Computational Linguistics, Machine Learning, NLP, Machine Translation
- **BSc Computer Science (1st Class, Hons)**
 - Nottingham Trent University, 2009-2013
 - Modules: Computer Security, Databases, Software Engineering, Distributed Network Architectures and Systems, Formal Reasoning, Foundations of AI.

Safety Projects

- **ToxScreen: Detecting Whether an LLM Has Been Poisoned**
 - Accepted ICML Mechanistic Interpretability Workshop 2026
 - Now under review @ AAAI 2027
 - Built a benchmark of ~800 backdoored models to for trigger recovery under realistic attacker/defender assumptions.
 - <https://arxiv.org/pdf/2607.26849>
- **Boundary-targeted Membership Inference Attacks on Safety Classifiers**
 - Under review at NeurIPS 2027
 - Designed membership inference attacks that target decision boundaries of safety classifiers, revealing privacy leakage in the models meant to keep LLMs safe.
 - <https://arxiv.org/abs/2605.22373>

Safety Experience

- **BlueDot: Technical AI Safety | May 2026 - June 2026 |**
- **Safety Fellowship | March 2026 - May 2026 |**
 - Mentors: Andrew Draganov (Arcadia/LASR Labs)
 - Project: Detecting Whether an LLM has Been Poisoned (accepted ICML mech interp workshop)

Selected Publications

- **PATCH: Mitigating PII Leakage in Language Models with Privacy-Aware Targeted Circuit Patching**
 - EACL 2026 | Findings
 - <https://arxiv.org/abs/2510.07452>
 - Located and edited the circuits responsible for PII memorisation in language models. Showed the leakage circuits survive existing defences, so the mechanism persists even where the behaviour appears suppressed.
 - **How Private are Language Models in Abstractive Summarisation?**
 - EMNLP 2025 | Main Proceedings
 - <https://aclanthology.org/2025.emnlp-main.1531/>
 - Ran the first systematic audit of privacy leakage in LM summarisation across six models, two prompting and fine-tuning regimes, and medical and legal corpora, with human evaluation.
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Internships

- **Vector Institute | June 2026 - August 2026 |**
 - Certified Unlearning in LLMs & Differentially Private Text Analysis
 - **JetBrains | March 2026 - July 2026 |**
 - Privacy preservation of code and proprietary information.
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Industry

- **Data Language | Jan 2014 - September 2023 | Junior Engineer → Lead Software Engineer → NLP Engineer**
 - Built a classification SaaS product, enabling clients easy access to text analytics. Led the development of a data viz tool allowing customers to view classification quality metrics.
 - **Ontoba | June 2013 - June 2014 | SWE**
 - Working on solving data silo issues with graphs/linked data focussed solutions.
 - **Press Association | July 2011- September 2012 | SWE (Intern)**
 - Working on a new digital platform centered around semantic web technologies.
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Mentoring Experience

- **Fatemeh Honarvar Nazari | Jan 2026 → Present**

Project: *Understanding how models reason about private information.*

- **Sajad Rahmanian Ashkezari, Neel Sanjaybhai Faganiya, Lucas Kopp | Sep 2025 → March 2026**

Project: *How Do Language Models Encode Privacy Norms?*

- **Yangming Cao | Feb 2025 -> Sept 2025**

Project: *Clinical Coding of Medical Texts*

- **Emma Ellwood | Apr 2025 -> Sept 2025**

Project: *Obfuscation of Gender and Familial Information in Medical Summaries*

Talks

- **EurIPS | December 2025 | Foundations of Language Model Security**

Title: *PATCH: Mitigating PII Leakage in Language Models with Privacy-Aware Targeted Circuit Patching*

- **EMNLP | November 2025 | Main Conference**

Title: *How Private are Language Models in Abstractive Summarization?*

- **Insigneo Institute | May 2024 | Synthetic Data Workshop**

Title: *Identifying and Aligning Medical Claims Made on Social Media with Medical Evidence*

Awards and Prizes

- **Best Poster, Insigneo Showcase, July 2025**
- **UKRI PhD Scholarship with the University of Sheffield, 2023-2027**